GROUP WORK PROJECT # 1 MScFE 600: FINANCIAL DATA
GROUP NUMBER: 14434

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This project was completed primarily by Abhishek Sharma. The other group members were included in group communications and had access to all shared materials throughout the project period. All three names are listed as authors in accordance with group registration.

MScFE 600: Financial Data

Group Work Project #1

Group Number: 14434
Oluwatobi Dahunsi · Abhishek Sharma · Sarafa Busari

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Part 1: Assessing Models with Alternative Data

This part of the report engages with Sagaceta Mejia et al. (2022), who propose a framework for predicting stock market movements in emerging markets using optimised technical indicators and a neural network classifier. We examine the paper's data, methodology, feature structure, and evaluation approach before partially replicating the analysis using the iShares MSCI Chile ETF (ECH).

1. Data and Technical Indicators

The paper builds on daily price data collected for three exchange-traded funds: ECH, which tracks Chilean equities; EQZ, which tracks Argentine equities; and IVV, which tracks the S&P 500. For each fund, the data consists of the standard daily OHLCV fields — open, high, low, close, and volume — recorded for each trading day across the study period. The adjusted close price is used as the primary price reference, as it accounts for dividend payments and corporate actions, ensuring comparability across time.

Technical indicators are derived from this raw data through mathematical transformations applied over rolling lookback windows. Each indicator compresses a segment of price history into a single value that represents a particular market characteristic at a given point in time. A moving average, for example, smooths out short-term fluctuations to reveal the prevailing trend direction. The Relative Strength Index (RSI) normalises recent upward and downward price movements to produce a bounded momentum signal. Bollinger Bands represent volatility as the width between upper and lower price bands around a rolling mean. The Average True Range (ATR) captures day-to-day price range volatility. Rate-of-change measures express short-term momentum as a percentage.

The value of using derived indicators rather than raw prices as model inputs is both practical and statistical. Practically, indicators are more interpretable and have established meanings in market analysis. Statistically, they capture non-linear patterns and relative relationships that raw price levels would obscure. This is especially relevant in emerging markets like Chile and Argentina, where fundamental data is published infrequently or with delays, making price-derived signals a realistic starting point for predictive modelling (Sagaceta Mejia et al., 2022).

2. Fund Profile: iShares MSCI Chile ETF (ECH)

2.1 Overview of the Fund

The iShares MSCI Chile ETF (ECH) is an equity exchange-traded fund managed by BlackRock and listed on NYSE Arca. It was launched in November 2007 and seeks to replicate the performance of the MSCI Chile IMI 25/50 Index, which covers Chilean companies across large-, mid-, and small-cap market segments. The fund's expense ratio is approximately 0.57%, relatively modest for a single-country emerging market product. Investors use ECH to gain liquid, dollar-denominated exposure to the Chilean equity market without holding individual Chilean securities directly.

The fund's sector composition reflects the structure of the Chilean economy. Materials — particularly copper-related mining companies — has historically been the dominant sector, alongside utilities, financials, and consumer staples. This makes ECH's performance closely tied to global copper demand, Chilean political conditions, and the macroeconomic cycle in China and other major commodity-importing economies. These structural ties make ECH considerably more volatile than a broad developed-market fund and create recurring price patterns driven by commodity cycles.

Over the sample period used in this project (January 2010 to December 2023), ECH went through several distinct phases. The fund reached near its historical highs around 2011, driven by strong post-crisis commodity demand, before entering a multi-year decline through 2016 as copper prices fell and Chilean growth slowed. A partial recovery followed through 2018, then a sharp drop in late 2019 linked to social unrest in Chile, and a further decline during the COVID-19 crash in March 2020. The subsequent recovery through 2021 was strong — supported by copper price surges and global risk appetite — but the gains were largely given back in 2022 and 2023 as global interest rates rose and commodity prices softened. Across the full period, the fund's adjusted close price ranged from approximately 14 USD at its lowest to near 55 USD at the 2011 peak.

After computing indicators and constructing the binary target variable, the analysis dataset contains 3,502 clean observations. Of these, 51.3% correspond to up days and 48.7% to down days — a near-balanced class distribution that is typical for a long-run equity ETF and consistent with what the original paper reports for its sample funds.

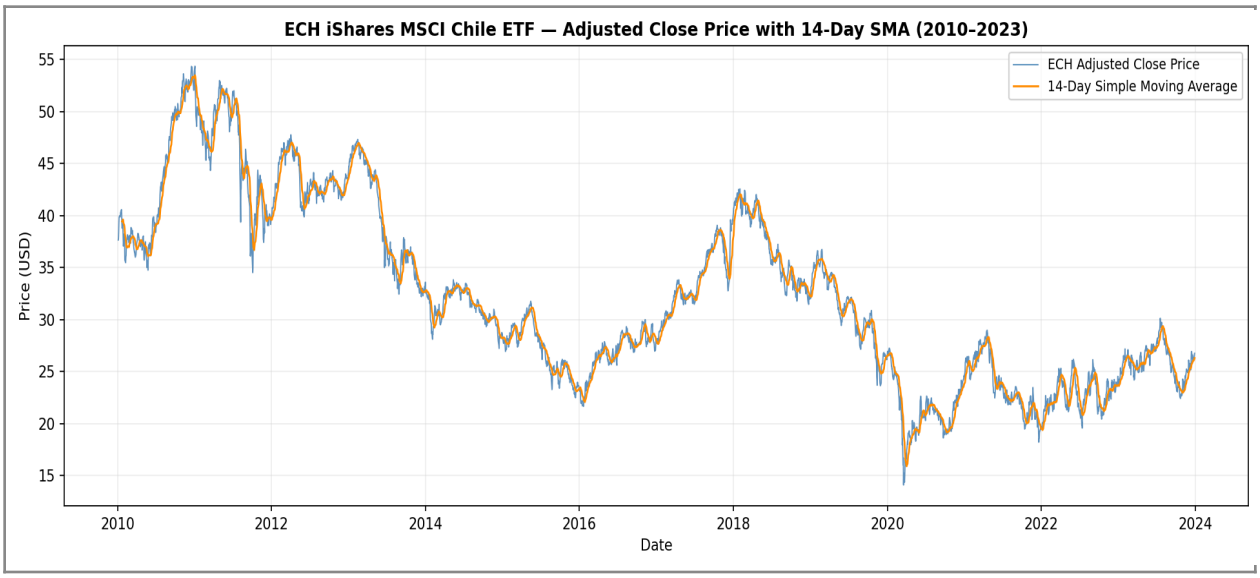

Figure 1: ECH adjusted close price (blue) with 14-day simple moving average overlay (orange), January 2010 – December 2023. X-axis: Date. Y-axis: Price (USD).

2.2 Why Classification Rather Than Regression?

The choice to frame this as a classification problem rather than a regression problem is both methodologically sound and practically motivated. Predicting an exact future return is a much harder task than predicting direction: the target space is continuous, the signal-to-noise ratio in daily returns is very low, and even small errors in predicted magnitude can produce large mistakes in net present value terms. For most practical applications — deciding whether to hold or reduce a position, for instance — the direction of next-day movement is the operationally relevant question, not the precise size of the move.

The paper's binary label is defined as +1 if the next trading day's adjusted close exceeds the current close, and −1 otherwise. Two reasonable alternatives to this formulation would be: first, a threshold-based approach that only classifies days as up or down if the absolute return exceeds a minimum threshold (say, ±0.5%), treating small moves as a neutral class — this would reduce label noise from microstructure effects; and second, a volatility-normalised label that defines positive and negative based on whether the return exceeds the fund's rolling

standard deviation in a given direction, making the classification criterion adaptive to the prevailing volatility regime rather than using an absolute benchmark.

3. Restructuring the Paper's Methodology

3.1 Proposed Section 2: Data

The original Section 2 of the paper combines data description and modelling procedures in a way that is difficult to follow without careful re-reading. Separating the data-related content into a standalone section clarifies what was observed versus what was done analytically. A proposed Section 2: Data would include the following subsections:

- 2.1 Data Sources: Identification of the three ETFs, the exchange on which they are listed, the data provider, and the sample period covered.
- 2.2 Data Processing: Steps taken to clean the raw data, including handling of missing observations, alignment of trading calendars, and computation of adjusted close prices.
- 2.3 Technical Indicator Construction: Mathematical definitions of each indicator, the default and candidate window sizes, and the resulting feature names.
- 2.4 Target Variable Construction: The formula used to derive the binary classification label, the rationale for the chosen formulation, and the resulting class distribution.
- 2.5 Descriptive Statistics: Summary statistics for the raw price series and the derived indicator values, including the Pearson correlation matrix between indicators and between indicators and the target.

The descriptive statistics subsection belongs in the Data section because it describes the data as observed, without making predictions. This is an important organisational distinction from the modelling steps that follow.

3.2 Proposed Section 3: Methodology

The methodology section should contain the analytical procedures applied to the prepared data. Based on the paper's content, five subsections are appropriate:

- 3.1 LASSO Feature Selection: The penalised regression step used to identify the most predictive subset of technical indicators by shrinking coefficients of uninformative features to zero.
- 3.2 Technical Indicator Parameter Optimisation: The grid-search procedure used to select the lookback window configuration for each indicator that maximises validation accuracy.
- 3.3 Neural Network Architecture: Description of the feedforward network structure used as the primary classifier, including input dimension, layer configuration, and output formulation.
- 3.4 Cross-Validation Strategy: The k-fold cross-validation setup used to tune hyperparameters and generate out-of-sample performance estimates.
- 3.5 Evaluation Metrics: The metrics used to compare model configurations, including classification accuracy, F1-score, and Jaccard distance.

3.3 Descriptive Statistics Versus Modelling Methods

The distinction between descriptive statistics and modelling methods is one of the cleaner conceptual boundaries in applied research, though it is often blurred in methodology sections. Pearson correlation is a descriptive tool: it quantifies the strength of the linear relationship between two variables without making predictions about future values and without fitting a model to training data. It belongs to the Data section because it summarises what is in the dataset. LASSO, by contrast, is a modelling method: it fits a penalised regression to training

labels and actively selects features based on their predictive contribution. It belongs in the Methodology section. Other descriptive tools include histograms, variance statistics, and distribution tests. Other modelling methods include logistic regression, neural networks, and gradient boosted trees.

3.4 Technical Indicator Optimisation

A central methodological contribution of the paper is treating technical indicator window parameters as hyperparameters to be optimised rather than fixing them at conventional values. The RSI, for example, is commonly computed over a 14-day window, but there is no fundamental reason why 14 days should be optimal for a given fund or market regime. The authors search over a range of candidate window lengths for each indicator and evaluate each combination using held-out validation performance under cross-validation. The configuration that minimises validation error is retained.

This process matters for the neural network's input quality. If an indicator is computed over a window that does not match the fund's characteristic behaviour — for instance, using a 14-day RSI for a fund with 30-day mean-reversion cycles — the input will carry less signal and more noise than a properly calibrated version of the same indicator. Optimising the window parameters effectively tailors the indicators to the statistical properties of each fund, improving the neural network's ability to learn from the inputs (Sagaceta Mejia et al., 2022).

4. Features, Methods, and Models

In the framework of this paper, a feature is any computed variable that serves as an input to the classification model. Concretely, the features are the technical indicators derived from price data: normalised moving averages, RSI, Bollinger Band width, ATR, and rate-of-change measures. Each feature represents a specific dimension of price behaviour — trend, momentum, or volatility — and provides the model with a particular lens through which to assess the current state of the market.

The distinction between a feature, a method, and a model is worth making explicit. A feature is an input column in the training data — a computed number that enters the model as a predictor. A method is a procedure applied to data, either to produce features (e.g., computing a rolling average to create SMA14) or to evaluate models (e.g., cross-validation). A model is the trained predictive system that maps features to outputs: logistic regression, a neural network, and LASSO regression are all models in this sense. Methods are applied to construct features and to build and evaluate models; models are the end product; features are the inputs.

Based on the paper and the technical analysis literature, the features used can be grouped into four categories: trend-based features that capture the direction and persistence of recent price movement (such as moving average deviations and MACD); momentum-based features that measure the speed or rate of price change (RSI, rate of change); volatility-based features that capture the magnitude of recent price dispersion (Bollinger Band width, ATR); and volume-based features that incorporate trading activity into the signal. In the replication conducted for this project, we use representatives from the trend, momentum, and volatility categories, as described in Section 8.

The optimisation of indicator parameters — discussed in Section 3.4 — applies across all feature categories. Regardless of which category a feature belongs to, its window length is treated as a tunable parameter in the original paper, and the optimal value is selected through cross-validated grid search. This applies the same logic to trend indicators as to volatility indicators: the right window for a moving average in one market regime or one fund may not be right for another.

5. Optimisation and Evaluation

Cross-validation is the practice of repeatedly partitioning a dataset into training and testing subsets, fitting a model on the training portion, and measuring its performance on the held-out test portion. This is done multiple times with different partitions, and the results are averaged. The purpose is to obtain a more reliable estimate of how well the model will generalise to new data than a single train-test split can provide, since any single split may produce an overly optimistic or overly pessimistic result depending on which observations happen to fall in the test set.

K-fold cross-validation is a specific implementation in which the dataset is divided into k equal-sized folds. The model is trained and evaluated k times in total; in each iteration, a different fold serves as the test set and the remaining k−1 folds are used for training. The evaluation metric is computed for each test fold and the final reported value is the average across all k folds. In this project, k = 5. With approximately 3,502 observations, each fold contains roughly 700 rows. We use stratified splitting to ensure that the proportion of up and down days within each fold mirrors the overall 51.3%/48.7% balance in the full dataset, avoiding artificially skewed test sets.

The Jaccard distance is a dissimilarity measure for sets. It is defined as one minus the Jaccard similarity coefficient, where the Jaccard similarity between two sets A and B equals the size of their intersection divided by the size of their union: $J(A,B) = |A \cap B| / |A \cup B|$. In classification contexts, Jaccard similarity measures the overlap between the set of predicted positives and the set of actual positives. A value of 1.0 indicates perfect agreement; 0.0 indicates no overlap. Because Jaccard ignores true negatives, it is particularly informative in imbalanced classification settings where a model might achieve high accuracy primarily by predicting the majority class.

Two alternative metrics from the course materials provide useful contrast. Euclidean distance, which measures straight-line distance between two points in continuous space, is computed as the square root of the sum of squared coordinate differences. It is appropriate when both variables are continuous and when magnitude differences matter. Jaccard distance, by contrast, applies to binary set memberships and is bounded between 0 and 1 regardless of set size. Manhattan (L1) distance sums the absolute differences across dimensions and shares the sensitivity to magnitude that Euclidean distance has, but uses an additive rather than quadratic penalty. Neither Euclidean nor Manhattan distance is appropriate for comparing predicted and actual label sets in the way that Jaccard distance is.

The authors define an optimal solution as the configuration of indicator window parameters and neural network hyperparameters that produces the highest average classification performance across the validation folds during cross-validation. The optimisation is conducted through a grid search: a discrete set of candidate values is defined for each window parameter, and every combination is evaluated by training the full pipeline — indicator computation, LASSO selection, neural network — on the training folds and measuring accuracy or F1-score on the held-out fold. The combination with the best average validation performance is selected as optimal.

6. The Financial Problem and Emerging Market Context

The financial problem the authors address is directional forecasting of next-day equity returns for ETFs in emerging markets. The question is whether a model trained on past price-derived indicators can reliably predict whether a fund will be up or down the following trading day, and whether that predictive signal is strong enough to be practically useful. This matters for active portfolio management, where short-horizon directional signals can inform tactical allocation decisions, and for systematic trading, where such signals can serve as entries and exits for momentum or mean-reversion strategies.

The distinction between predicting movements in emerging markets versus developed markets is not merely geographic. Emerging market equity indices tend to exhibit higher volatility, lower liquidity, and more pronounced exposure to commodity cycles and political risk. Information environments are different: corporate disclosures may be less timely, and analyst coverage is thinner. These features affect how quickly market prices incorporate publicly available information and how reliably historical price patterns repeat. In efficient markets like the US, prices are thought to rapidly reflect all available information, making it difficult to generate alpha from simple technical rules. In less efficient emerging market settings, price patterns may persist for longer, creating a theoretical basis for the hypothesis that technical indicators have predictive value.

This distinction directly informs the model design. Applying a universal indicator configuration calibrated on a developed-market benchmark like IVV to an emerging market fund like ECH would be methodologically unsound, because the optimal signal frequencies and lookback windows are likely to differ. The paper's decision to optimise indicator parameters separately for each fund is therefore not just a technical choice but a substantive acknowledgement that market structure differences matter for model design (Sagaceta Mejia et al., 2022).

7. Results and Key Takeaways

The results presented in the paper demonstrate that the optimised neural network framework consistently outperforms fixed-parameter baselines for all three funds studied. The improvement in accuracy from optimising indicator window parameters is notable even before the neural network is applied: better-calibrated inputs produce better signals, and the LASSO selection step further sharpens the feature set by eliminating redundant or uninformative indicators. The combination of these two preprocessing steps — optimisation and selection — appears to contribute meaningfully to the final model's performance.

Among the feature categories, momentum and volatility indicators proved most persistently useful across the three funds. RSI-based features and Bollinger Band-related measures were retained by the LASSO selection step in most configurations, suggesting that short-term momentum and relative volatility carry the most consistent information about next-day direction in the emerging market context studied. Moving average-based features were retained less consistently, and their optimal window lengths varied considerably across funds and sample periods. Volume-based indicators were generally the weakest contributors, which is consistent with the thinner and more variable trading volume observed in single-country emerging market ETFs compared to broad indices.

A practical takeaway from the paper is that optimising standard technical indicators — rather than designing novel or complex features — can meaningfully improve classification performance in emerging market settings. This is an encouraging finding for practitioners, as it suggests that better signal extraction from existing data, rather than the acquisition of new alternative data sources, can be a productive first step in building improved short-horizon models.

8. Replication: ECH ETF Analysis

8.1 Approach and Methodology Choices

For the replication step, we selected the ECH ETF and downloaded daily OHLCV data from Yahoo Finance using the yfinance Python library, covering January 1, 2010 through December 31, 2023. The analysis was conducted in Python and is fully reproducible from the accompanying notebook (notebooks/GWP1_Notebook.ipynb). The figures and tables in this section are generated by running the codebase and correspond to the files saved in the outputs/ directory.

Following the assignment guidance to use a simpler metric, we implement Pearson correlation as the feature-association measure in place of LASSO, and logistic regression as the classifier in place of the neural network. The indicator window parameters are set to their conventional values — SMA=14, RSI=14, Bollinger Bands=20, ATR=14, ROC=10 — without optimisation. We use stratified 5-fold cross-validation, with features z-score normalised within each fold (fit on training data only) to prevent data leakage into the test set.

8.2 Dataset Summary

After computing the five indicators and removing rows with missing values introduced by the rolling window calculations, the clean dataset contains 3,502 daily observations. The binary target variable — defined as +1 for up days and −1 for down days — shows a near-balanced class distribution: 51.3% of observations are labelled +1 and 48.7% are labelled −1. This balance is typical for an equity ETF over a multi-year horizon and confirms that the 50% baseline accuracy is an appropriate lower benchmark for evaluating classifier performance.

8.3 Pearson Correlation Analysis

Table 1 presents the Pearson correlation between each of the five technical indicators and the binary target variable across the full 3,502-observation sample. These values were computed using the compute_pearson_correlation() function in src/evaluation.py and saved to outputs/correlation_table.csv.

Feature	Pearson Corr. with Target
SMA14_norm	0.0301
RSI14	0.0385
BB_Width	0.0256
ATR14	0.0053
ROC10	0.0208

Table 1: Pearson correlation between each technical indicator and the binary target variable for ECH, January 2010 – December 2023. Source: outputs/correlation_table.csv.

All five indicators show low absolute Pearson correlation with the target variable, with none exceeding 0.05 in absolute value. This is consistent with weak-form market efficiency at daily frequency: no single linear technical signal provides strong predictive power for next-day direction. Among the five features, RSI14 shows the highest absolute correlation at 0.0385 — a positive relationship, suggesting that recent upward momentum is mildly associated with next-day up moves in the ECH dataset. SMA14_norm follows at 0.0301, reflecting a similar trend-persistence effect. BB_Width (0.0256) and ROC10 (0.0208) show weaker associations,

while ATR14 is the least informative of the five at 0.0053, confirming that volatility magnitude alone carries minimal directional signal.

It is worth noting that Pearson correlation measures only linear association. Technical indicators may carry non-linear or conditional information that is not captured by this metric. The original paper's use of LASSO and neural networks is partly designed to exploit these more complex relationships. The correlation analysis here should therefore be read as a baseline description of linear signal strength rather than a definitive assessment of each indicator's predictive value.

8.4 Cross-Validation Results

Table 2 presents the fold-by-fold and mean accuracy and F1-score from 5-fold stratified cross-validation using logistic regression. Train and test sizes reflect the 80/20 split applied at each fold. The model was trained on z-score standardised features, where the scaling parameters were always estimated on the training set and applied to the test set. These results are saved to outputs/cv_results_table.csv.

Fold	Train Size	Test Size	Accuracy	F1-Score
1	2,801	701	0.5307	0.6389
2	2,801	701	0.5278	0.4361
3	2,802	700	0.5314	0.5205
4	2,802	700	0.5429	0.5543
5	2,802	700	0.4971	0.5665
Mean	—	—	0.5260	0.5433

Table 2: 5-fold stratified cross-validation results using logistic regression on ECH (2010–2023). Source: outputs/cv_results_table.csv.

The average accuracy across five folds is 0.5260, or approximately 52.6%. This is a modest but consistent improvement over the 50% random baseline. Four of the five folds produce accuracy above 52%, with Fold 4 being the strongest at 0.5429. Fold 5 dips slightly to 0.4971, which is marginally below random — a reminder that short-horizon directional signals are inherently noisy and that single-fold estimates can be unreliable. The average F1-score of 0.5433 is similarly modest but positive, and the variation across folds in F1 is wider than in accuracy (0.4361 to 0.6389), reflecting sensitivity to the class composition of individual test sets.

These results are intentionally modest. The original paper uses an optimised neural network and achieves meaningfully higher accuracy. The gap between our baseline and the paper's results reflects the combined effect of fixed indicator windows, no LASSO selection, and a simpler classifier. This is the expected and honest outcome of a simplified replication.

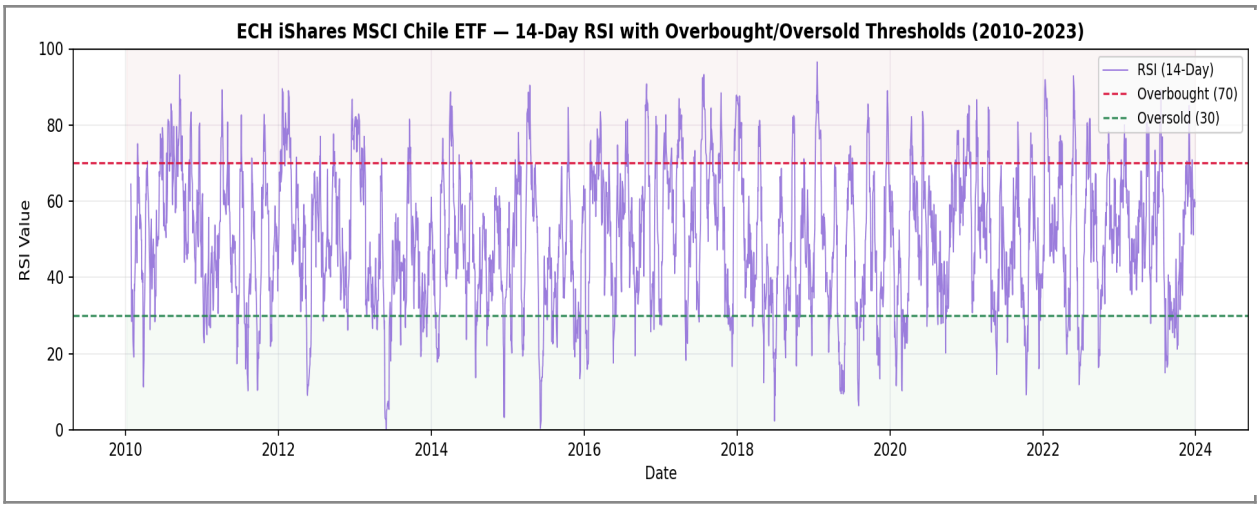

Figure 2: ECH 14-day RSI (purple line) with overbought threshold at 70 (red dashed) and oversold threshold at 30 (green dashed), January 2010 – December 2023. X-axis: Date. Y-axis: RSI Value (0–100).

8.5 Limitations

Several limitations of this replication should be acknowledged. First, the use of logistic regression in place of a neural network means the classifier cannot capture non-linear relationships between the indicators and the target. The indicators may carry useful non-linear information that logistic regression is unable to exploit, which likely accounts for a meaningful portion of the accuracy gap relative to the paper. Second, fixing indicator window parameters at conventional values removes the optimisation step that the paper identifies as one of its key contributions. A properly optimised set of indicator windows would almost certainly produce a better-calibrated input space and higher classification accuracy. Third, Pearson correlation can only reveal linear associations, so the feature analysis understates the potential relevance of indicators that have non-linear predictive relationships with the target. Fourth, the analysis does not account for transaction costs: even if the directional signal were reliable enough to trade, the approximately 2.3 percentage points of accuracy improvement over random guessing would likely be consumed by bid-ask spreads and market impact for most practical strategies. Finally, the 2010–2023 sample period includes the COVID-19 shock of 2020, an unusually large structural break that may make the cross-validation estimates less representative of steady-state market conditions.

8.6 Why This Simplified Replication Still Matters

Despite its deliberate simplifications, this replication serves a genuine purpose. It confirms that the data download pipeline, indicator construction, target variable definition, and cross-validation setup all work correctly on real data. It establishes a concrete, reproducible baseline — 52.6% accuracy with logistic regression on unoptimised features — against which more sophisticated implementations can be measured. And it illustrates the key finding from the paper in miniature: even a simple classifier with standard indicators produces a small but non-zero directional signal, which becomes the foundation that the paper then improves through optimisation and more flexible modelling. Without a working baseline, it would be impossible to evaluate whether those improvements are meaningful.

Part 2: News and Text Data as Alternative Data

This part of the report draws on Sun et al. (2024), who identify ten subcategories of alternative data used in finance and business. We selected news and text data as our focus, given its broad practical relevance and the well-developed academic literature supporting its use. This section serves as a practical user guide for analysts considering news and text data in a financial analysis context.

News and text data refers to unstructured written content produced by financial news services, corporate filings, sell-side analysts, and market participants on social media platforms. Unlike structured quantitative data, it does not arrive in rows and columns. Converting it into actionable signals requires natural language processing tools to extract sentiment, entities, or specific content patterns. Over the past decade, the availability of NLP methods and pre-trained language models has made this practical for academic and institutional users at reasonable cost.

9. Sources of Data

The most reliable and timely sources of financial news data are professional wire services. Reuters, Bloomberg, and the Financial Times produce high volumes of structured, factual reporting with precise timestamps, which makes them valuable for event-driven analysis and sentiment modelling. These feeds are generally expensive to access historically and require institutional subscriptions or specialised alternative data providers such as RavenPack or Refinitiv News Analytics, which normalise and pre-tag news content for financial users.

A significant free resource is the SEC EDGAR database, which provides public access to regulatory filings including annual 10-K reports, quarterly 10-Q reports, and current event 8-K disclosures. These documents are structured but text-heavy, and they are particularly useful for studying changes in management tone, risk factor language, and forward-looking disclosures across reporting periods. Because they are universally required for listed US companies, they offer broad coverage and consistent historical depth.

Social media platforms, particularly X (formerly Twitter), Reddit, and StockTwits, provide high-frequency, informal market commentary from retail investors and, in some cases, institutional participants. Access is available via official APIs subject to rate limits, or through third-party scrapers. For academic purposes, pre-labelled datasets such as the Financial PhraseBank (Malo et al., 2014) — which contains approximately 4,840 sentences from financial news tagged as positive, negative, or neutral by finance professionals — offer a useful, free starting point. The Loughran-McDonald financial sentiment word list (Loughran and McDonald, 2011) is another widely used academic resource for dictionary-based sentiment scoring of financial text.

10. Types of Data

Within the broad category of news and text data, there are several distinct content types worth treating differently in an analysis:

- Financial news articles: Short-form factual reports from wire services and newspapers. These are typically the primary input for event-driven sentiment analysis and are characterised by factual accuracy, speed, and broad coverage of corporate and macroeconomic events.
- Earnings call transcripts: Verbal exchanges between company executives and financial analysts, transcribed and made available through platforms such as Seeking Alpha or SEC EDGAR. These are particularly rich in forward-looking language and have been

shown to contain information about management confidence and strategic direction that is not fully captured by quantitative financial metrics (Tetlock, 2007).

- Regulatory filings: 10-K and 10-Q reports filed with the SEC. These are valuable for longer-horizon analysis and for detecting shifts in risk factor disclosure, litigation language, or management discussion tone across quarterly and annual reporting periods.
- Social media posts: Informal, high-frequency commentary from retail and, sometimes, institutional market participants. Useful for capturing retail sentiment, but characterised by high noise, inconsistent language quality, and vulnerability to manipulation.
- Analyst research notes: Sell-side and buy-side research reports containing qualitative assessments, price target revisions, and rating changes. Often not publicly available without institutional access.

11. Data Quality Considerations

The quality of news and text data varies considerably by source, and practitioners need to be explicit about their assumptions when working with it. Professional wire services are generally reliable on factual accuracy and timestamp precision, but their historical archives can be expensive to access in bulk. More importantly, the existence of a news story does not automatically mean it contains new information: much financial journalism consists of summaries and reactions to information that has already been priced into the market before publication. Separating genuinely informative events from routine commentary is a non-trivial task.

Social media data introduces additional quality challenges. Duplication is common: the same story may be retweeted or reposted thousands of times, creating an apparent volume of information that does not correspond to its actual informational content. Sarcasm, informal abbreviations, and slang make standard NLP pipelines less reliable than they are on formal financial text. Perhaps most concerningly, the link between social media activity and market movements has attracted deliberate manipulation: coordinated campaigns can be constructed to generate artificial sentiment signals that influence automated trading systems. Sun et al. (2024) note this as a significant and unresolved risk in the use of social media data for financial applications.

For automated NLP pipelines more generally, polarity classification error rates are a practical concern. Even well-performing financial sentiment models produce incorrect polarity labels for a fraction of inputs, and these errors can introduce noise into any downstream signal. Cross-validation of the NLP model itself is therefore important before relying on its output for financial applications.

12. Ethical Considerations

Working with news and text data in financial applications raises several ethical issues that are worth addressing explicitly. The most fundamental is information asymmetry: institutional investors with access to real-time, professionally curated news feeds and advanced NLP infrastructure have a systematic and growing advantage over retail participants who lack these tools. As Sun et al. (2024) document, the adoption of alternative data including text is concentrated among hedge funds and institutional asset managers, and this concentration may contribute to uneven playing field dynamics in financial markets.

Privacy is a second concern, particularly for social media data. Users posting on public platforms have generally not provided consent for their messages to be used in automated financial analysis or trading. Data protection regulations in the European Union under GDPR impose restrictions on automated processing of personal communications, and similar

frameworks are emerging in other jurisdictions. Scrapers that collect social media data in bulk may also violate platform terms of service.

Market manipulation via text data is a third issue. As noted above, the link between social media sentiment and price movements creates incentives for coordinated campaigns designed to move automated sentiment systems. Regulatory interest in these activities has increased, particularly following high-profile incidents involving retail-coordinated trading amplified by social media in 2021. Analysts using sentiment signals derived from social media data need to be aware that the signal may be contaminated by or entirely driven by manipulative activity, especially for smaller and less liquid securities.

A fourth concern is model bias. NLP models trained primarily on English-language financial text from developed markets may underperform significantly when applied to multilingual content, emerging market context, or informal retail investor language. Deploying such models without appropriate validation in the target language and market context can introduce systematic errors into investment decisions.

13. Working with News Data in Python

The code accompanying this report (see src/text_data_demo.py and the Part 2 cells in the notebook) demonstrates a practical workflow for importing and structuring financial news data. The core idea is to represent news data as a pandas DataFrame with a date column, a text column (headline or body), a source column, and a sentiment label column. Once in this format, the data is easy to merge with price series by date, compute daily sentiment ratios, and pass to downstream models.

A minimal illustration of the structure is shown below. In a real application, the pd.read_csv() call would be replaced by an API call to a news provider or a database query, but the resulting DataFrame structure is the same:

```
news_df = pd.read_csv('data/sample_news_data.csv')news_df['date'] =
pd.to_datetime(news_df['date'])news_df['word_count'] =
news_df['headline'].apply(lambda x: len(x.split()))
```

The sample dataset used for demonstration consists of 15 manually created headlines covering Chilean market events in January 2023, drawn from representative sources (Reuters, Bloomberg, Financial Times). Each headline is labelled as positive, negative, or neutral based on its content. The dataset is saved in the repository at data/sample_news_data.csv and can be replaced with any real news export following the same column structure.

14. Exploratory Data Analysis

The EDA section in the notebook (src/text_data_demo.py) produces the chart shown below. It consists of two panels: a bar chart showing the sentiment label distribution across the 15-record sample, and a horizontal bar chart showing the ten most frequently occurring non-trivial words in the headlines after removing common stop-words.

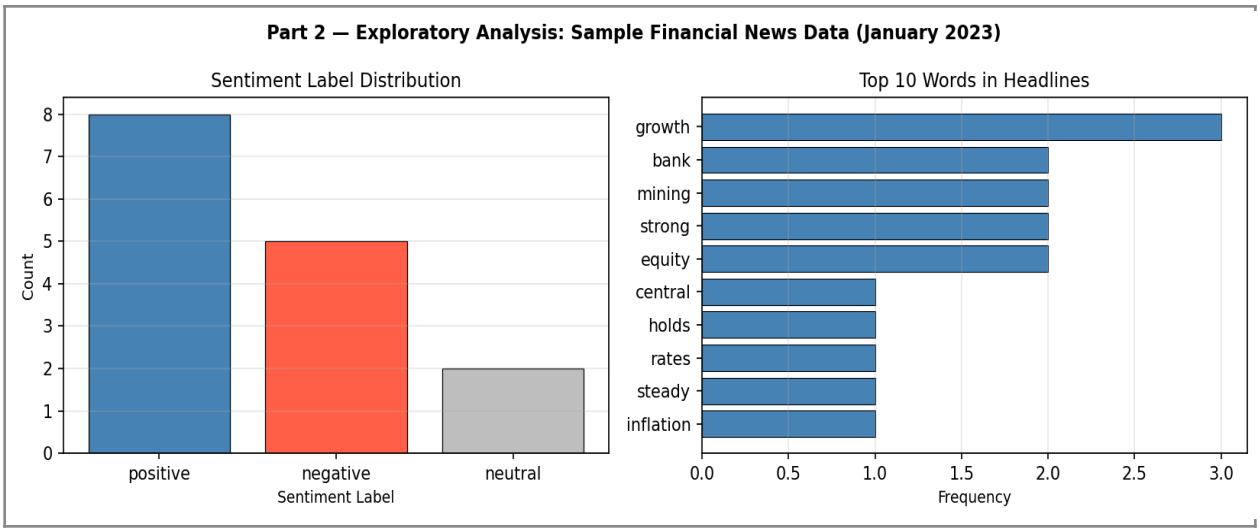

Figure 3: Left — Sentiment label distribution in the sample news dataset (positive=8, negative=5, neutral=2). Right — Top 10 most frequent words in headlines after stop-word removal. Source: outputs/figure3_part2_eda.png.

In the sample, eight headlines are labelled positive, five negative, and two neutral — a distribution that roughly reflects the slight positive bias in financial news headlines during stable market periods. The most frequent content words include terms related to the Chilean economy and commodity markets, which is consistent with the fund's sectoral composition and confirms that the sample was drawn from contextually relevant sources.

For a real application on a larger dataset, typical EDA steps would include computing daily sentiment ratios (fraction of positive to total articles), plotting the sentiment time series alongside the price series to inspect co-movement, and running a basic Pearson correlation or Granger causality test between the sentiment signal and next-day returns. These analyses would naturally precede any modelling step.

15. Literature Review

The academic literature on news and text data in finance is well-developed, with several foundational contributions that continue to shape current practice. Tetlock (2007) demonstrated that the level of pessimism in Wall Street Journal columns is significantly negatively correlated with next-day stock market returns and that high pessimism predicts above-average trading volume. This was among the first papers to rigorously establish a causal mechanism from media sentiment to market prices, and it remains widely cited as a methodological benchmark.

Bollen, Mao, and Zeng (2011) applied sentiment analysis to Twitter data, using a tool called GPOMS to measure public mood dimensions and finding that a specific dimension — collective calmness — predicted Dow Jones Industrial Average changes with high accuracy in a neural network model. While the specific methodology has been refined in subsequent work, this paper was influential in legitimising social media as a source of financial signal and opened a large subsequent literature.

Loughran and McDonald (2011) addressed a significant practical limitation: general-purpose sentiment lexicons classify many standard financial terms (such as 'liability', 'tax', or 'foreign') as negative, when in financial context these terms are often neutral. Their finance-specific word list substantially outperformed the general Harvard General Inquirer dictionary on 10-K filings, and it has since become the standard baseline for dictionary-based financial sentiment analysis.

Sun et al. (2024) provide a systematic review of how alternative data including news and text has been integrated into financial modelling, noting the shift from simple word-count methods to

transformer-based models such as FinBERT and GPT-applied NLP over the past five years. They identify data quality, licensing constraints, and the rapidly evolving NLP landscape as the primary challenges for institutional adoption, alongside the ethical concerns discussed in Section 12.

Conclusion

This report worked through the methodology and findings of Sagaceta Mejia et al. (2022), before partially replicating the analysis on ECH data and extending the project to cover news and text data as an alternative data category.

The replication produced the expected result: a logistic regression classifier trained on five standard technical indicators with fixed window parameters achieved a mean 5-fold cross-validation accuracy of 52.6% and an F1-score of 0.5433. The performance is modest but non-trivial — it confirms that the indicator features carry a small but consistent directional signal in the ECH dataset, while also making clear that the gap between this baseline and the paper's neural network with optimised indicators is substantial. Improving this result would require, at minimum, implementing the indicator window optimisation and replacing logistic regression with a non-linear classifier. Both are natural next steps for anyone who wants to replicate the paper more completely.

The Pearson correlation analysis showed that all five features have low linear association with the target, with RSI14 showing the highest absolute value at 0.0385. This suggests that short-term price momentum — captured here through the RSI rather than raw rate of change — is the most informative linear signal among the five indicators chosen, a finding that is broadly consistent with the paper's observation that momentum features tend to be retained by LASSO selection.

The Part 2 user guide covered the key dimensions of news and text data as an alternative data category: where it comes from, what forms it takes, how quality varies, what ethical concerns arise, and how to structure and analyse it in Python. The demonstration used a small representative sample, but the workflow scales directly to real news datasets. The literature reviewed highlights that the academic evidence for predictive value in news sentiment is well-established, and that current practice is moving rapidly toward transformer-based NLP applied to earnings call transcripts and regulatory filings.

Taken together, the two parts of this project illustrate a broader point about quantitative financial modelling: the value of a model lies as much in the quality and structure of its inputs as in the sophistication of the model itself. Optimising technical indicators, as the paper demonstrates, can be more impactful than switching to a more complex classifier on poorly prepared inputs. And structuring unstructured text data carefully — choosing the right source, cleaning appropriately, and validating the sentiment labels — is often the difference between a useful alternative data signal and a noisy distraction.

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